Co-Design Rankings Do Not Survive Realization: An 18-Morphology, 4-Environment Study

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***Abstract*—Learning-based co-design methods score candidate robot bodies in a simplified simulator and return a ranked list. We ask how much of that ranking survives when the designs are made buildable. We take 18 morphologies of a modular icosahedral legged robot, sampled from our own evolutionary co-design space, and train each with a matched PPO recipe in a replica of the abstract simulator used by the search and in a fabricable simulator calibrated against CAD models and motor datasheets. The two rankings are uncorrelated ($\rho_s = +0.11$, $p = 0.66$; mean rank change 5.9 of 18): the best abstract design drops to 7th, and a design that barely moved in the abstract simulator (0.54 m, 17th) becomes 2nd at 10.8 m. The reshuffle is robust to the training seed ($P < 0.02$ under resampling), to reward re-weighting and added mass ($\rho_s \geq 0.77$) and to halving the training budget ($\rho_s \geq 0.90$), and it is caused mainly by the changes in geometry, DoF allocation, rest pose and mass rather than by the actuator model ($\rho_s = -0.30$ vs. $+0.61$). Changing the terrain at fixed realization, in contrast, largely preserves the ranking (rubble $\rho_s = 0.90$, ice 0.72); only stepping stones reshuffle it as much. We trace the reshuffle to a shell-sliding exploit, to rest poses that penetrate or float above the ground, and to a leg-count advantage that exists only in the abstract simulator. Simulator fidelity should therefore be part of the co-design loop rather than a step after it.**

I. Introduction

Most learning-based co-design methods, whether they propose morphologies generatively or improve them iteratively by RL, optimization or evolution [1]–[6], produce the same kind of output: a ranked shortlist of bodies, each scored by a policy trained in a deliberately simplified simulator. Our own prior work follows this pattern: ICOS is an orientation-free Platonic shell in the tradition of polyhedral robots [7]–[11], and we searched its leg placements with a speciated evolutionary algorithm in MuJoCo MJX [12] (seven DoF-partition species). Here we do not repeat that search; we sample 18 bodies from its design space and ask what happens to their ranking when each body becomes a machine printed and assembled from Dynamixel XM430-W350 servos.

II. Experimental Setup

Robot and morphologies. ICOS is a 20-face icosahedral shell (160 mm circumdiameter) to which identical 3-DoF leg modules are bolted on chosen faces. All 20 faces are equivalent, so a morphology is a subset of faces (60,249 for 3–6 legs), and its symmetry follows from where and how the legs are mounted. The realized quadruped weighs 2.20 kg, stands 134 mm tall and uses 12 servos. We study 4 hand-designed baselines and 14 generation-0 samples, two per species; the set is a stratified sample of the search space, not of its best designs.

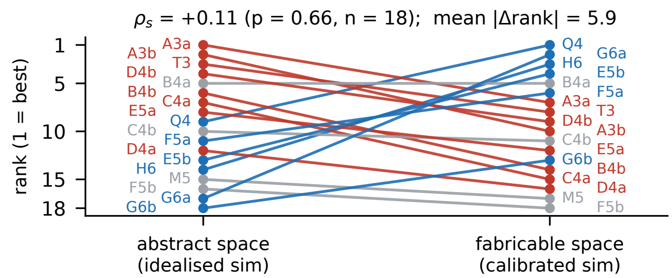

Fig. 1. Flat-ground ranking of the 18 morphologies in the abstract (left) and fabricable (right) space. Blue lines rise, red lines fall, grey lines move at most 2 ranks. Q4/T3/M5/H6: hand-designed quadruped, tripod, mixed, hexapod; A–G: the seven DoF-partition species, two generation-0 samples (a, b) each; the digit is the leg count.

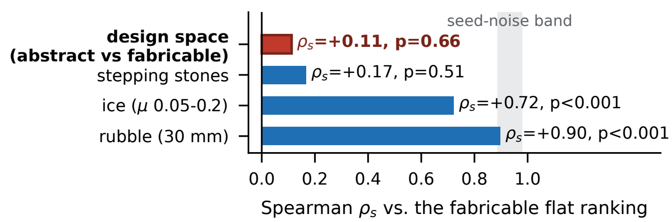

Fig. 2. Spearman rank correlation of each ranking with the fabricable flat-ground ranking. Shaded: the 5–95% range that seed noise alone produces at the median measured CV.

Training. All runs use a matched recipe: Genesis [13] GPU simulation for both design spaces (the abstract space re-implements the search's MJX models), 4096 parallel environments, PPO (rsl-rl [14]), 2000 iterations (197M steps), 20 s episodes, and 50 Hz control on top of 250 Hz physics. The reward is $r = 2.5\,\Delta d + 0.5\,\text{alive} - 0.01\|a\|^2$; in the fabricable space a thermal penalty of -0.1 is added, except in the ice runs (Sec. IV). The only other difference between the two spaces is the entropy coefficient (0.01 abstract, 0.003 fabricable). We report the mean displacement over 4096 rollouts (never the best of several automatic retries) and an *honest* variant that truncates each rollout at its first fall (Sec. III-C).

Realization. Five things change between the abstract and the fabricable model. (1) The freely partitioned 12-DoF budget (2–5 DoF per leg) becomes a fixed printable 3-DoF module, so DoF $= 3\times$legs; a $(4, 4, 4)$ design, for example, cannot be built from one module. (2) Sphere collision proxies become a convex-hull body mesh with motor cylinders. (3) The straight-leg rest pose, which the real robot cannot reach, becomes a solved static equilibrium ($\Delta z < 25$ mm, no penetration). (4) The ideal position servo becomes a torque PD controller limited by a torque–speed envelope (4.1 N·m stall, 4.8 rad/s no-load), with 0–20 ms latency, sensor noise and the thermal penalty. (5) Lumped masses become CAD part masses at their centroids, so mass now scales with leg count (1.8–3.1 kg and 9–18 servos for 3–6 legs). Changes (2)–(5) follow standard sim-to-real practice [15], [16], with parameters taken from CAD over seven hardware generations.

III. RESULTS

A. Realization reshuffles the ranking

The two flat-ground rankings (Fig. 1) are statistically unrelated ($\rho_s = +0.11$, $p = 0.66$); the mean rank change is 5.9 of 18, close to the 6.0 expected for independent rankings. The best abstract design (3 legs, 17.5 m) finishes 7th at 5.53 m, and two designs that barely moved in the abstract space (0.54 and 0.13 m) finish 2nd (10.8 m) and 13th.

Seed noise does not explain this result. Retraining 6 morphologies with 3 seeds each gives a median coefficient of variation of 0.13 in the fabricable space (range 0.03–0.63) and 0.24 in the abstract space (0.02–0.39); seed-to-seed rank agreement is $\rho_s = 0.77$–0.89 and 0.83–0.89. Under the null hypothesis that both spaces observe one unchanged ranking through their own seed noise, 20,000 resamples give $P(\rho_s \leq 0.11) < 10^{-4}$ at the median noise levels and 0.019 at the highest.

Reward weights and training budget do not explain it either. Raising the distance weight of the abstract reward from 1.0 to 2.5, or adding 57% realistic part mass, changes the abstract ranking little ($\rho_s = 0.77$–0.97) without moving it toward the fabricable one ($\rho_s \leq 0.11$); the rankings at half the training budget already match the final ones ($\rho_s \geq 0.90$) and already disagree across spaces ($\rho_s = 0.21$).

To separate the actuator model from the other changes, we retrained the 18 fabricable bodies with the ideal servos of the abstract space. The resulting ranking is unrelated to the abstract ranking ($\rho_s = -0.30$, $p = 0.23$) but close to the fully calibrated one ($\rho_s = +0.61$, $p = 0.007$). The changes in geometry, DoF allocation, rest pose and mass therefore account for most of the reshuffle. The actuator model perturbs the ranking further and unevenly (the best design loses 13%, one design loses 55%, another gains 2.5×), which is why 0.61 lies below the seed-noise band at median noise (0.89–0.98).

B. The task environment mostly does not

With the fabricable model fixed, we retrained all 18 morphologies on quantized fractal rubble (30 mm), ice ($\mu \in [0.05, 0.2]$) and stepping stones (0.25 m stones, 50 mm gaps over a 0.5 m drop). Rank correlation with the flat-ground ranking remains high (Fig. 2): $\rho_s = 0.90$ for rubble, inside the median-noise seed band, and 0.72 for ice, below it (both $p < 0.001$; the ice runs lack the thermal term, Sec. IV). Only stepping stones reshuffle the ranking as much as realization does ($\rho_s = 0.17$, $p = 0.51$). There, 15 of 18 morphologies learn to survive (standing still is valid), but only 4 of 18 travel more than 1 m (median 0.55 m).

C. Why the ranking changes

The reshuffle is not a scoring artifact: re-scoring every rollout at its first fall leaves both rankings unchanged (no abstract policy ever falls; $\rho_s = 0.94$ between the two scorings and still 0.07 against the fabricable ranking; 17 of 18 fabricable policies agree within 6%). We find three causes. *(i) Simulation exploits disappear.* The tripod, 3rd in the abstract space with 11.1 m, achieved this by sliding its smooth spherical shell along the ground, which sphere collision proxies allow and a convex-hull mesh does not; in the fabricable space it walks instead and reaches 5.46 m (8th). *(ii) The rest pose matters.* Of the 18 original home keyframes, 6 penetrated the ground and 3 floated above it. Re-solving them (morphology and reward unchanged) changed flat-ground displacement from 5.33 to 10.84 m and from 5.60 to 10.09 m for two designs and from 8.44 to 5.79 m for a third. Eight of the nine were re-trained with the re-solved pose; the ninth (D4b, floating) keeps its original run. *(iii) The abstract space rewards a trade-off the hardware cannot make.* In the abstract space, displacement is predicted by leg count: fewer, longer legs win ($\rho_s = -0.89$). After realization this relation disappears ($\rho_s = +0.07$), because the fixed module replaces the trade-off between legs and DoF with one between legs and mass. Mirror symmetry of the leg placement, our search's leading hypothesis, shows no detectable relation to displacement in either space at this sample size ($|\rho_s| \leq 0.40$, $p \geq 0.10$); only two designs, the tripod and one 5-leg sample, have a mirror-symmetric placement at all.

D. Advantage depends on the environment

No morphology dominates across environments. On ice, median retention (environment / flat displacement) increases from 3 to 5 legs (0.33, 0.81, 1.06) and is 0.88 for 6 legs (possibly inflated by the missing thermal term); rubble costs little (0.93) and stepping stones almost everything (0.12).

IV. DISCUSSION

A shortlist scored in an abstract simulator says little about the order of the same designs after realization, but it does transfer across tasks. Since iterative search and zero-shot generation share this scoring step, simulator fidelity belongs inside the co-design objective.

Limitations. Runs outside the two 6×3 replications use a single seed, and abstract-space values are the noisier ones. The main result survives the highest measured seed noise in both spaces ($P = 0.019$); the finer comparisons (actuator model, ice) hold only at median noise (the worst-case band, 0.22–0.82, contains both). One reward inconsistency remains: the ice runs omit the thermal term, so part of the ice result may reflect reward rather than terrain. In the servo experiment, geometry, DoF allocation, rest pose and mass remain confounded. The abstract space replicates the MJX search environment in Genesis, unchecked against it; no hardware results yet.

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SUPPLEMENTARY MATERIAL

This appendix collects the full per-morphology results, the training and realization parameters, the seed replications behind the resampling null, and the secondary analyses referred to in the two-page abstract. Every number is computed from the same evaluation files as the main text (mean displacement over 4096 rollouts of $20\,\mathrm{s}$ unless stated otherwise); the scripts that produce the tables are in the repository (`paper_iros2026/`).

APPENDIX A
THE 18 MORPHOLOGIES

Fig. 3 shows every morphology at its rest pose in both design spaces, ordered by abstract flat-ground rank. Table I lists the displacement of each morphology in every condition of the study. Labels: Q4/T3/M5/H6 are the hand-designed quadruped, tripod, mixed and hexapod; A–G are the seven DoF-partition species of the evolutionary search, a/b the two generation-0 samples of each; the digit is the leg count. F and J are the icosahedral edit distances of the leg placement to the nearest mirror-symmetric placement (faces only, and faces with motor pattern); 0 means exactly mirror-symmetric.

APPENDIX B
TRAINING RECIPE BY RUN FAMILY

All runs use Genesis GPU simulation, 4096 parallel environments, PPO (rsl-rl), 2000 iterations (197M environment steps), $20\,\mathrm{s}$ episodes, $50\,\mathrm{Hz}$ control on top of $250\,\mathrm{Hz}$ physics, and the reward $r = 2.5\,\Delta d + 0.5\,\mathrm{alive} - 0.01\|a\|^2$. Table II lists the terms that differ between run families, read back from the stored configuration of every run ("calibrated" = torque PD with envelope, latency, noise and domain randomization; "ideal" = the abstract position servo). The thermal term penalizes the mean commanded torque above the continuous rating ($1.4\,\mathrm{N{\cdot}m}$); it is defined only when the actuator model is active, so it is identically zero in runs without it (abstract space, ideal-servo split), whether or not it appears in the configuration. The ice runs were trained without it by oversight (Sec. IV). The two extra penalties of the first baseline stepping-stone runs (torque reversal and touchdown, -0.02 each) were removed by re-running those four morphologies; the paper uses the re-runs.

TABLE II
REWARD TERMS AND ENTROPY COEFFICIENT PER RUN FAMILY.

Run family	Thermal -0.1	Entropy	Actuator
Abstract, flat	inactive	0.01	ideal
Abstract, seeds 2–3	inactive	0.01	ideal
Fabricable, flat / rubble	yes	0.003	calibrated
Fabricable, ice	**no**	0.003	calibrated
Fabricable, stones (all 18)	yes	0.003	calibrated
Fabricable, seeds 2–3	yes	0.003	calibrated
Ideal-servo split (18)	inactive	0.003	ideal

APPENDIX C
REALIZATION PARAMETERS AND REST-POSE RE-SOLUTION

Table III gives the parameters of the fabricable model. Randomized quantities are re-sampled per environment at every reset. Table IV lists the nine morphologies whose original home keyframe penetrated the ground or floated above it, with their displacement before and after the rest pose was re-solved (same morphology, reward and training recipe). Eight of the nine were re-trained with the re-solved pose and those runs are used throughout the paper; the ninth (D4b, a floating pose) was not re-trained before submission and keeps its original run ("–").

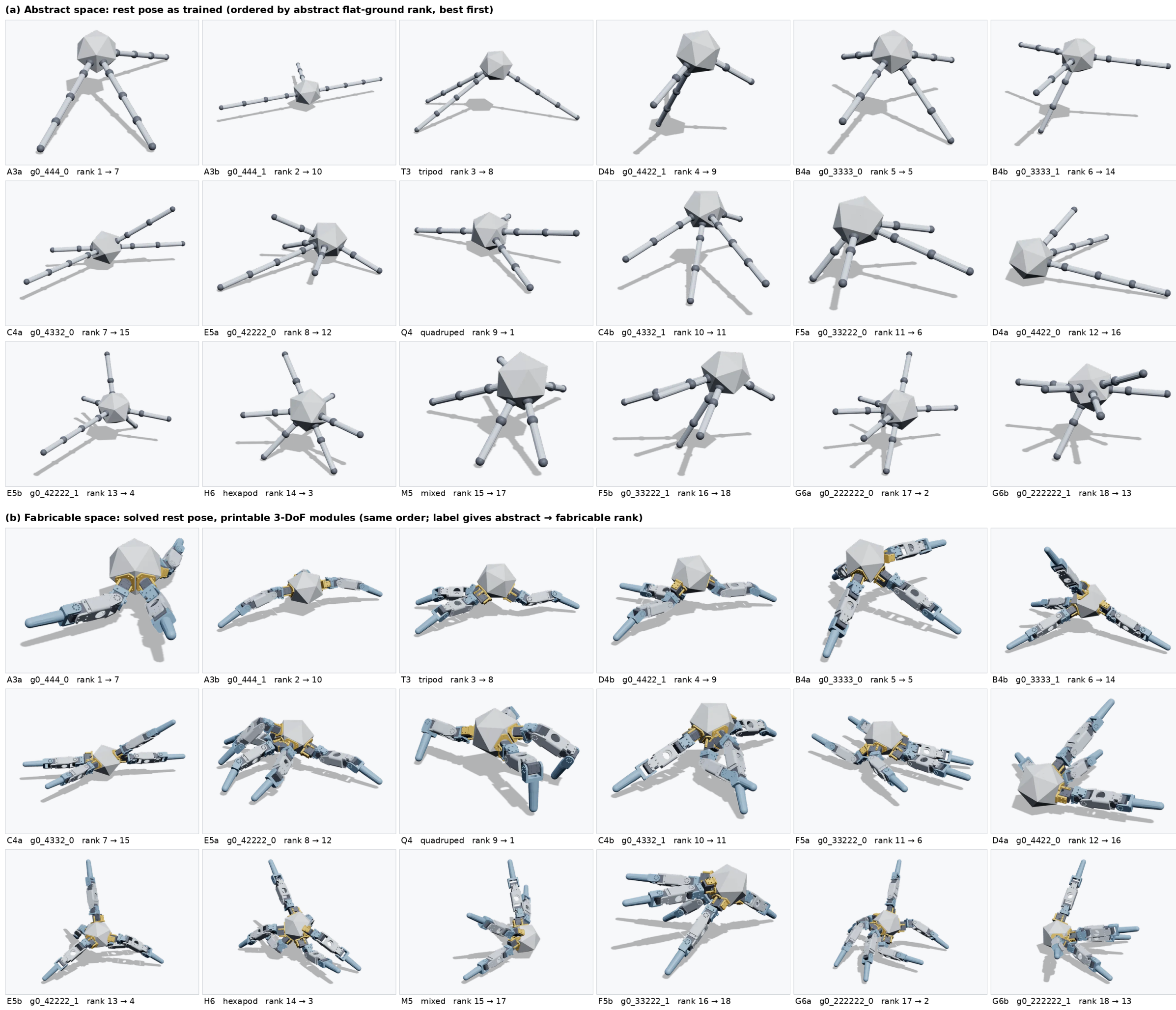

Fig. 3. The 18 morphologies at their rest poses. (a) Abstract space: sphere-proxy bodies, freely partitioned DoF, straight-leg rest pose. (b) Fabricable space: convex-hull shell, printable 3-DoF leg modules, solved static rest pose. Ordered by abstract flat-ground rank; each label gives the abstract → fabricable rank.

TABLE III
FABRICABLE-MODEL PARAMETERS (DYNAMIXEL XM430-W350 AT 12 V).

Quantity	Value
Leg module	3 DoF, identical for every leg
Body collision	convex hull of the shell + motor cylinders + skirt rings
Rest pose	solved static equilibrium; gate $\Delta z < 25\,\mathrm{mm}$, $\Delta q < 0.1\,\mathrm{rad}$, no penetration
Servo model	explicit torque PD, $k_p \sim \mathcal{U}[25, 60]$, $k_v \sim \mathcal{U}[0.5, 1.2]$
Torque–speed envelope	$\tau_{\max} = 4.1\,(1 - \lvert\dot{q}\rvert/4.8)\,\mathrm{N{\cdot}m}$, scale $\sim \mathcal{U}[0.8, 1.05]$
Command latency	0–5 physics substeps (0–20 ms), per episode
Observation noise	joint pos. 0.005 rad, joint vel. 0.2 rad/s, ang. vel. 0.05 rad/s, gravity 0.02
Velocity quantization	0.024 rad/s (Dynamixel resolution); gyro bias std 0.02 rad/s
Joint damping	0.15 N·m·s/rad (abstract space: 0.85)
Ground friction	$\mu \sim \mathcal{U}[0.25, 1.25]$ per environment (ice runs: $[0.05, 0.2]$)
Pushes	every 4 s, horizontal velocity impulse $\mathcal{U}[0.1, 0.3]$ m/s, random direction
Masses	CAD part masses at centroids: 1.8–3.1 kg and 9–18 servos for 3–6 legs

TABLE IV
DISPLACEMENT (M) BEFORE / AFTER RE-SOLVING THE REST POSE, FABRICABLE SPACE.

Morph.	Original pose	Flat	Rubble	Ice
H6	penetrating	5.60 / 10.09	4.61 / 9.25	6.63 / 8.89
C4b	penetrating	0.56 / 4.80	4.06 / 5.19	0.08 / 6.16
D4a	penetrating	2.34 / 2.26	1.65 / 2.71	2.21 / 1.83
F5a	penetrating	7.18 / 5.64	6.96 / 6.94	8.05 / 7.49
F5b	penetrating	2.30 / 1.47	2.67 / 1.18	2.31 / 1.65
G6a	penetrating	5.33 / 10.84	4.87 / 9.62	3.60 / 8.95
A3a	floating	5.42 / 5.53	4.47 / 4.52	2.99 / 1.42
B4a	floating	8.44 / 5.79	5.86 / 5.25	7.22 / 3.93
D4b	floating	5.30 / –	5.17 / –	6.49 / –

APPENDIX D
SEED REPLICATIONS AND THE RESAMPLING NULL

Six morphologies (the fabricable flat-ground winner, two mid-ranked and three low-ranked designs) were trained with three seeds in each space; seed 1 is the run used in the main tables (Tables V and VI). The coefficient of variation (CV) is the sample standard deviation over seeds divided by the mean. Seed-to-seed rank agreement over the six morphologies

TABLE I

MEAN 20 S DISPLACEMENT (M) OF EACH MORPHOLOGY IN EVERY CONDITION. ABSTRACT: REPLICA OF THE SEARCH SIMULATOR. IDEAL SERVO: FABRICABLE BODY WITH THE ABSTRACT SERVO MODEL (SEC. III-A). FABRICABLE: CALIBRATED MODEL ON FLAT GROUND, RUBBLE, ICE AND STEPPING STONES. RANKS ARE FOR FLAT GROUND.

Morph.	Species	Legs	Abstract	rank	Ideal servo	Fabricable	rank	Rubble	Ice	Stones	F	J
Q4	(3,3,3,3) h.-d.	4	7.75	9	13.50	11.72	1	9.70	8.50	0.39	2	6
G6a	(2,2,2,2,2,2)	6	0.54	17	12.25	10.84	2	9.62	8.95	0.54	2	4
H6	(2,2,2,2,2,2) h.-d.	6	3.39	14	11.88	10.09	3	9.25	8.89	1.93	2	4
E5b	(4,2,2,2,2)	5	4.11	13	10.43	9.45	4	8.52	7.15	1.12	1	2
B4a	(3,3,3,3)	4	8.93	5	2.29	5.79	5	5.25	3.93	0.36	1	3
F5a	(3,3,2,2,2)	5	7.27	11	6.94	5.64	6	6.94	7.49	4.02	0	5
A3a	(4,4,4)	3	17.53	1	3.68	5.53	7	4.52	1.42	0.63	2	8
T3	(4,4,4) h.-d.	3	11.12	3	5.85	5.46	8	4.06	3.57	0.48	0	0
D4b	(4,4,2,2)	4	9.80	4	3.49	5.30	9	5.17	6.49	0.45	1	8
A3b	(4,4,4)	3	12.39	2	6.42	5.24	10	4.99	1.73	0.55	1	4
C4b	(4,3,3,2)	4	7.51	10	8.16	4.80	11	5.19	6.16	0.58	1	4
E5a	(4,2,2,2,2)	5	7.84	8	9.87	4.44	12	3.58	4.72	2.55	2	6
G6b	(2,2,2,2,2,2)	6	0.13	18	7.07	4.39	13	5.39	6.44	0.18	3	6
B4b	(3,3,3,3)	4	8.82	6	6.79	3.31	14	3.75	5.67	0.33	1	3
C4a	(4,3,3,2)	4	8.45	7	3.05	2.72	15	2.96	1.12	0.39	2	6
D4a	(4,4,2,2)	4	5.90	12	4.18	2.26	16	2.71	1.83	0.41	2	6
M5	(3,3,2,2,2) h.-d.	5	3.26	15	3.09	2.13	17	2.22	1.39	0.57	2	6
F5b	(3,3,2,2,2)	5	2.80	16	0.46	1.47	18	1.18	1.65	0.77	1	5

is $\rho_s = 0.77$, 0.77 and 0.89 in the fabricable space and 0.83, 0.89 and 0.83 in the abstract space.

Null model. Under the hypothesis that the two spaces rank the morphologies identically and differ only by seed noise, we draw $20{,}000$ pairs of rankings: the abstract displacements are multiplied by independent factors $1+\epsilon$, $\epsilon \sim \mathcal{N}(0, \mathrm{CV})$, once with the abstract CV and once with the fabricable CV, and the Spearman correlation between the two perturbed rankings is recorded. The observed $\rho_s = 0.11$ is reached in no draw at the median CVs (0.24 / 0.13; $P < 10^{-4}$) and in 1.9% of the draws at the largest measured CVs (0.39 / 0.63; $P = 0.019$). With a single fabricable CV applied to one ranking (the convention of the first version of this study) the corresponding values are $P < 10^{-4}$ and 0.006.

Seed-noise band (Fig. 2). The shaded band is the 5–95% range of ρ_s between the fabricable flat-ground ranking and a copy of it perturbed by the median fabricable CV (0.89–0.98). At the largest fabricable CV the band widens to 0.22–0.82, which is why the ice (0.72) and ideal-servo (0.61) comparisons are stated at the median only.

TABLE V

FABRICABLE FLAT GROUND, THREE SEEDS (M).

Morph.	seed 1	seed 2	seed 3	CV
F5b	1.47	1.37	0.28	0.63
B4a	5.79	3.38	3.61	0.31
E5b	9.45	8.91	9.06	0.03
C4b	4.80	5.96	5.01	0.12
A3b	5.24	4.69	5.23	0.06
Q4	11.72	10.11	9.00	0.13
median CV				0.13

TABLE VI

ABSTRACT SPACE, THREE SEEDS (M).

Morph.	seed 1	seed 2	seed 3	CV
F5b	2.80	2.93	4.36	0.26
B4a	8.93	7.41	7.46	0.11
E5b	4.11	6.28	6.28	0.23
C4b	7.51	3.25	5.58	0.39
A3b	12.39	12.68	12.82	0.02
Q4	7.75	4.95	8.82	0.28
median CV				0.24

APPENDIX E
REWARD WEIGHTS AND TRAINING BUDGET

Four abstract-space variants were trained for all 18 morphologies: distance weight 1.0 (the search's reward), 1.5, 2.5 (the value used in the main text) and 2.5 with 57% added part mass at the joints. Table VII gives the rank correlations between them and with the fabricable flat-ground ranking. The ranking at half the training budget is compared with the final ranking through the training-curve return at iteration 1000 versus 2000: $\rho_s = 0.91$ in the fabricable space and 0.91 in the abstract space, and the two spaces already disagree at half budget ($\rho_s = 0.21$). The median change of the curve over the last fifth of training is 4%.

TABLE VII

SPEARMAN ρ_s BETWEEN ABSTRACT-SPACE REWARD VARIANTS ($n = 18$) AND WITH THE FABRICABLE FLAT-GROUND RANKING.

	$d=1.0$	$d=1.5$	$d=2.5$	$d=2.5$, $+57\%$ m.	vs. fabr.
$d=1.0$	–	0.96	0.78	0.77	-0.07
$d=1.5$	0.96	–	0.78	0.78	-0.07
$d=2.5$	0.78	0.78	–	0.77	+0.07
$d=2.5$, $+57\%$ mass	0.77	0.78	0.77	–	+0.11

APPENDIX F
ENVIRONMENTS AND LEG-COUNT RETENTION

Rubble is a quantized fractal heightfield with 30 mm steps; ice is flat ground with friction $\mu \sim \mathcal{U}[0.05, 0.2]$; stepping stones are 0.25 m square platforms separated by 50 mm gaps over a 0.5 m drop. Retention is the displacement in the environment divided by the flat-ground displacement of the same morphology. Fig. 4 and Table VIII give the median per leg-count group; the ice values may be inflated by the missing thermal term.

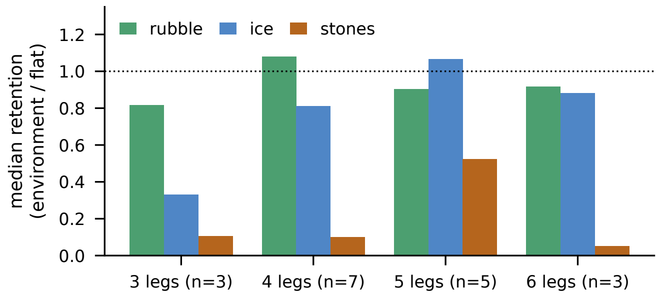

Fig. 4. Median retention (environment / flat displacement) per leg-count group.

TABLE VIII
MEDIAN RETENTION PER LEG-COUNT GROUP AND OVER ALL 18 MORPHOLOGIES.

Environment	3 legs ($n=3$)	4 legs ($n=7$)	5 legs ($n=5$)	6 legs ($n=3$)	all
rubble	0.82	1.08	0.90	0.92	0.93
ice	0.33	0.81	1.06	0.88	0.82
stones	0.11	0.10	0.52	0.05	0.12

APPENDIX G
FIRST-FALL ("HONEST") SCORING

Mean displacement over 4096 rollouts is the primary score. The honest variant truncates each rollout at its first fall (body orientation beyond the reset threshold) and reports the displacement at that moment. Table IX compares the two scorings. None of the 18 abstract policies ever falls; the abstract exploits are slides and hops, not falls, so honest and mean rankings agree ($\rho_s = 0.94$) and the honest abstract ranking is as unrelated to the fabricable one as the mean ranking ($\rho_s = 0.07$). In the fabricable space the honest evaluation was run on the campaign before the nine rest poses were re-solved; 17 of 18 policies agree within 6%, and the exception (C4b, a penetrating rest pose, fall rate 1.0) is one of the re-solved designs.

TABLE IX
MEAN VERSUS FIRST-FALL DISPLACEMENT (M). FABRICABLE VALUES ARE FROM THE PRE-RE-SOLUTION CAMPAIGN, IN WHICH THE HONEST EVALUATION WAS RUN; † MARKS THE NINE DESIGNS RE-SOLVED AFTERWARDS (TABLE IV).

	Abstract		Fabricable flat (pre-re-solution)		
Morph.	mean	honest	mean	honest	fall rate
Q4	7.75	7.31	11.72	11.79	0.00
T3	11.12	11.44	5.46	5.42	0.00
M5	3.26	3.59	2.13	2.12	0.01
H6†	3.39	3.46	5.60	5.55	0.01
A3a†	17.53	17.38	5.42	5.42	0.00
A3b	12.39	13.21	5.24	5.21	0.01
B4a†	8.93	10.11	8.44	8.44	0.00
B4b	8.82	8.84	3.31	3.15	0.14
C4a	8.45	7.91	2.72	2.71	0.01
C4b†	7.51	7.49	0.56	0.32	1.00
D4a†	5.90	6.15	2.34	2.33	0.00
D4b†	9.80	6.97	5.30	5.25	0.02
E5a	7.84	8.24	4.44	4.40	0.00
E5b	4.11	4.11	9.45	9.48	0.00
F5a†	7.27	7.18	7.18	7.21	0.00
F5b†	2.80	2.85	2.30	2.26	0.08
G6a†	0.54	0.55	5.33	5.20	0.02
G6b	0.13	0.13	4.39	4.36	0.00

APPENDIX H
SYMMETRY OF THE LEG PLACEMENT

The evolutionary search's working hypothesis was that mirror-symmetric leg placements walk better. We scored the placement of each morphology with seven definitions of attachment-pattern symmetry: the icosahedral edit distance to the nearest mirror-symmetric placement on faces only (F) and with motor pattern (J), and continuous symmetry measures (Zabrodsky et al., 1992) of the leg-direction set with respect to the nearest of the icosahedron's 15 mirror planes, any mirror plane, any 2-fold axis, the best of mirror / 2-fold / 3-fold, and the mirror measure standardized against random placements with the same leg count. Table X gives the Spearman correlation of each score with displacement in each space, oriented so that a positive value means "more symmetric walks farther". None is significant, and the signs change between definitions; at $n = 18$ only $|\rho_s| \geq 0.62$ would be detectable with 80% power. Only two placements are exactly mirror-symmetric (T3 and F5a, $F = 0$). These scores describe the placement pattern; whether a placement is a symmetry of the assembled robot also depends on how each leg module is oriented, which is the subject of ongoing work.

TABLE X
SPEARMAN ρ_s (AND p) BETWEEN SEVEN LEG-PLACEMENT SYMMETRY SCORES AND DISPLACEMENT, $n = 18$, ORIENTED SO THAT + MEANS MORE SYMMETRIC WALKS FARTHER.

Definition	Abstract	Fabricable flat
F: edit distance, faces	$+0.40$ ($p = 0.10$)	$+0.09$ ($p = 0.71$)
J: edit distance, faces + motors	-0.00 ($p = 0.99$)	$+0.28$ ($p = 0.25$)
CSM, nearest of 15 icosahedral mirror planes	$+0.06$ ($p = 0.82$)	$+0.10$ ($p = 0.69$)
CSM, nearest mirror plane (any)	$+0.31$ ($p = 0.20$)	-0.17 ($p = 0.50$)
CSM, nearest 2-fold axis (any)	-0.05 ($p = 0.83$)	$+0.23$ ($p = 0.37$)
CSM, best of mirror / 2-fold / 3-fold	$+0.31$ ($p = 0.21$)	-0.33 ($p = 0.19$)
CSM mirror, z-score given leg count	$+0.16$ ($p = 0.53$)	-0.11 ($p = 0.65$)

APPENDIX I
COMPUTE, HARDWARE LINE AND CODE

Each training run takes 0.4–1.4 GPU-hours on an Apple M5 Pro (Genesis, Metal backend). The paper rests on about 200 runs: 18 abstract and 4×18 fabricable rankings, 24 pre-resolution runs, 18 ideal-servo runs, 2×12 seed replications, 4 clean stepping-stone re-runs and 3×18 abstract reward variants, plus the first-fall evaluations. The fabricable model was calibrated over seven CAD generations of the same leg module (Fig. 5); the realized quadruped weighs $2.20\,\mathrm{kg}$ and stands $134\,\mathrm{mm}$ tall. Models, trajectories, evaluation files, the scripts that produce every table in this appendix, and an interactive viewer of the trained behaviours are released with the paper.

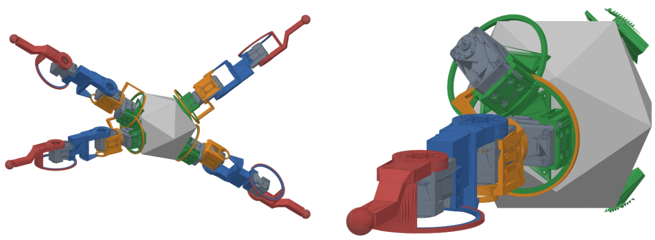

Fig. 5. CAD assembly of the fabricable quadruped (left) and one leg module (right): green, base mount on a shell face; orange and blue, the two printed servo connectors; red, the foot link. The same module is bolted to every chosen face.